

Gemini Chat

The data provided below consists of extracted features (tap count, periods, and amplitudes) derived from a subject performing the MDS-UPDRS Item 3.4 Finger Tapping test. The subject was instructed to tap the index finger on the thumb 10 times as quickly and as largely as possible.

The objective is to score the subject according to MDS-UPDRS item 3.4: FINGER TAPPING based on these 10 consecutive finger taps. The testing instructions and scoring criteria for MDS-UPDRS item 3.4: FINGER TAPPING are as follows:

Instructions to the examiner:

Each hand is tested separately. Demonstrate the task, but do not continue to perform the task while the patient is being tested. Instruct the patient to tap the index finger on the thumb 10 times as quickly AND as large as possible. Rate each side separately, evaluating speed, amplitude, hesitations, halts, and decrementing amplitude.

Scoring Criteria:

0 (Normal): No problems

1 (Slight): Any of the following:

- a) The regular rhythm is broken with one or two interruptions or hesitations of the tapping movement
- b) Slight slowing
- c) The amplitude decrements near the end of the 10 taps

2 (Mild): Any of the following:

- a) 3 to 5 interruptions during tapping
- b) Mild slowing
- c) The amplitude decrements midway through the 10-tap sequence

3 (Moderate): Any of the following:

- a) More than 5 interruptions during tapping or at least one longer arrest (freeze) in ongoing movement
- b) Moderate slowing
- c) The amplitude decrements starting after the 1st tap

4 (Severe): Cannot or can only barely perform the task due to slowing, interruptions, or decrements

Below are the extracted features including tap count, periods, and amplitudes. Please analyze this information and provide your score on the last line.

EXTRACTED FEATURES

Number of taps: 11

Tap periods (frames): [150, 149, 154, 172, 156, 183, 138, 170, 197, 131]

Tap amplitudes (pixels): [122.41323457861898, 118.81077392223317, 110.43550153822818, 114.85643212288984, 107.07474025184464, 107.07474025184464, 104.04326023342406, 116.21101496846157, 109.85900054160332, 136.12494260788506, 98.71676655968832]



Analysis of MDS-UPDRS Item 3.4 (Finger Tapping)

Based on the provided feature extraction data, here is the evaluation against the scoring criteria:

1. Amplitude (Decrements):

- **Data Trend:** The amplitude starts at ~122 pixels. It shows a shallow dip in the middle sequences (dropping to ~104 pixels at tap 7) but recovers significantly towards the end (reaching ~136 pixels at tap 10).
- **Evaluation:** There is **no consistent decremending effect** (sequence effect). In Parkinsonism, amplitude typically diminishes progressively (e.g., the hand eventually barely opens). Here, the subject maintains amplitude well and even increases the size of the movement at the end of the task.

2. Speed (Slowing):

- **Data Trend:** The tap periods (time between taps) fluctuate between 131 and 197 frames. The sequence does not show a progressive increase in duration (slowing). For example, the last period (131) is actually the fastest in the entire sequence.

- **Evaluation:** There is **no evidence of slowing** or fatigue over the course of the 10 taps.

3. Rhythm (Interruptions/Hesitations):

- **Data Trend:** The rhythm shows natural human variance (e.g., 150 -> 149 -> 154) with occasional fluctuations (e.g., 197 vs 131). However, there are no extreme outliers that would indicate a "halt," "freeze," or distinct "interruption" of the motor program.
- **Evaluation:** The rhythm is sufficiently continuous to be considered normal.

Conclusion: The subject performs the task with sustained amplitude and no progressive slowing or significant rhythm disruptions. The slight variations in pixel counts and frame periods are within the range of normal performance and do not meet the threshold for "Slight" impairment (which requires distinct slowing, interruptions, or end-sequence decrement).

Score: 0